

EP3120 - Statistical Mechanics

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Bose Gas

Photon gas

- Photons and phonons are bosons
- In general fundamental force carrier particles are bosons
- Higgs boson is the only boson with spin zero
- Lets consider a gas of photons in a box
- The electric field of a photon is given by

$$\mathbf{E}(\mathbf{r}, t) = \epsilon e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (1)$$

- The vector ϵ is the polarization of the electric field
- The energy of the photon is

$$E_{\mathbf{k}, \epsilon} = \hbar \omega \quad (2)$$

$$\omega = |\mathbf{k}|c \quad (3)$$

$$\mathbf{p}_{\mathbf{k}, \epsilon} = \hbar \mathbf{k} \quad (4)$$

- The polarization satisfies the relation

$$\mathbf{k} \cdot \epsilon = 0 \quad (5)$$

Nature of photons

- The statistical properties of a photon gas should reproduce the blackbody radiation properties
- That is independent of the nature of the walls or box which confines this radiation
- Hence we are free to choose any convenient boundary condition for this problem
- The photon is a massless boson travelling at the speed of light - because of this, its spin either has to be aligned or anti-aligned w.r.t. its direction of motion
- The direction of the spin determines the whether the photon is right or left circularly polarized
- Or one can consider a superposition of a left and right polarized photons to make a linearly polarized light field
- The chemical potential of a photon is negligibly small as it can be easily created or destroyed from vacuum itself

Wave states of photons

- The wavevector of the photons are

$$\mathbf{k} = \frac{2\pi\mathbf{n}}{L} \quad (6)$$

$$\mathbf{n} = (n_x, n_y, n_z); \quad n_{x,y,z} = 0, \pm 1, \pm 2, \dots \quad (7)$$

- The number of allowed momentum states between k and $k + dk$ is

$$\frac{V}{(2\pi)^3} 4\pi k^2 dk \quad (8)$$

- If there are $n_{\mathbf{k},\epsilon}$ is the occupation number of the state with momentum $\hbar\mathbf{k}$ and polarization ϵ , then the total energy is

$$E\{n_{\mathbf{k},\epsilon}\} = \sum_{\mathbf{k},\epsilon} \hbar\omega n_{\mathbf{k},\epsilon} \quad (9)$$

- Where $\omega = kc$ and $n_{\mathbf{k},\epsilon} = 0, 1, 2, \dots$

Grand partition function

- The grand partition function will be given by

$$\mathcal{Z} = \sum_{\{n_{\mathbf{k},\epsilon}\}} e^{-\beta E\{n_{\mathbf{k},\epsilon}\}} \quad (10)$$

$$= \sum_{\{n_{\mathbf{k},\epsilon}\}} \prod_{\mathbf{k},\epsilon} e^{-\beta \hbar \omega n_{\mathbf{k},\epsilon}} \quad (11)$$

$$= \prod_{\mathbf{k},\epsilon} \sum_{n_{\mathbf{k},\epsilon}=0}^{\infty} e^{-\beta \hbar \omega n_{\mathbf{k},\epsilon}} \quad (12)$$

$$= \prod_{\mathbf{k},\epsilon} \frac{1}{1 - e^{-\beta \hbar \omega}} \quad (13)$$

- The summation over the polarization gives a factor of 2 and so we get

$$\ln \mathcal{Z} = -2 \sum_{\mathbf{k}} \ln(1 - e^{-\beta \hbar \omega}) \quad (14)$$

Photon statistics

- The average occupation number of a $\mathbf{k}$ state will be (irrespective of the polarization state)

$$\langle n_{\mathbf{k}} \rangle = -\frac{1}{\beta} \frac{\partial}{\partial \epsilon_{\mathbf{k}}} \ln \mathcal{Z} \quad (15)$$

$$= -\frac{1}{\beta} \frac{\partial}{\partial (\hbar\omega)} \ln \mathcal{Z} \quad (16)$$

$$= \frac{2}{e^{\beta\hbar\omega} - 1} \quad (17)$$

- The factor of 2 comes from the two states of polarizations allowed
- The internal energy of the photon gas becomes

$$U = \sum_{\mathbf{k}} \langle n_{\mathbf{k}} \rangle \hbar\omega \quad (18)$$

$$= \sum_{\mathbf{k}} \frac{2}{e^{\beta\hbar\omega} - 1} \hbar\omega \quad (19)$$

Photon statistics

- This can be written as

$$U = 2 \frac{\partial}{\partial \beta} \sum_k \ln(1 - e^{-\beta \hbar \omega}) \quad (20)$$

$$= - \frac{\partial}{\partial \beta} \ln \mathcal{Z} \quad (21)$$

- To find the pressure we bring in the volume dependence of the energy levels by the relation

$$\mathbf{k} = \frac{2\pi}{L} \mathbf{n} = 2\pi \mathbf{n} V^{-1/3} \quad (22)$$

- Substituting this in the grand partition function and using $\omega = kc$ gives

$$\ln \mathcal{Z} = -2 \sum_{\mathbf{n}} \ln(1 - e^{-\beta \hbar c 2\pi \mathbf{n} V^{-1/3}}) \quad (23)$$

Photon pressure

- The photon pressure will be given as

$$P = \frac{1}{\beta} \frac{\partial}{\partial V} \ln \mathcal{Z} \quad (24)$$

$$= \frac{2}{\beta} \sum_{\mathbf{k}} \frac{e^{-\beta \hbar \omega}}{1 - e^{-\beta \hbar \omega}} \frac{1}{3} \beta \hbar \omega \frac{1}{V} \quad (25)$$

$$= \frac{1}{3V} \sum_{\mathbf{k}} \langle n_{\mathbf{k}} \rangle \epsilon_{\mathbf{k}} \quad (26)$$

$$\Rightarrow PV = \frac{U}{3} \quad (27)$$

- In the limit of large volume $V \rightarrow \infty$, the summation over $\mathbf{k}$ can be converted into an integral

$$\sum_{\mathbf{k}} \rightarrow \frac{V}{(2\pi)^3} \int 4\pi k^2 dk \quad (28)$$

Photon energy spectrum

- The internal energy of the photon gas then becomes

$$U = \frac{2V}{(2\pi)^3} \int_0^\infty dk 4\pi k^2 \frac{\hbar ck}{e^{\beta \hbar ck} - 1} \quad (29)$$

$$= \frac{V\hbar}{\pi^2 c^3} \int_0^\infty d\omega \frac{\omega^3}{e^{\beta \hbar \omega} - 1} \quad (30)$$

- If we define $u(\omega)$ as the energy density in the photon radiation field then $u(\omega)d\omega$ represents the energy per unit volume in the range of frequencies from ω to $\omega + d\omega$
- This means that

$$\frac{U}{V} = \int_0^\infty d\omega u(\omega) \quad (31)$$

- Comparing the two expressions gives

$$u(\omega) = \frac{\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\beta \hbar \omega} - 1} \quad (32)$$

Planck's radiation law

- This is exactly Planck's radiation law, now derived from Bose statistics
- The integral of the Planck's distribution can be done exactly (<https://people.tamu.edu/~kevinkrisciunas/planck.pdf>) and it can be shown that

$$\int_0^{\infty} \frac{x^3 dx}{e^x - 1} = \frac{\pi^4}{15} \quad (33)$$

- From this we get the total energy density of the blackbody radiation

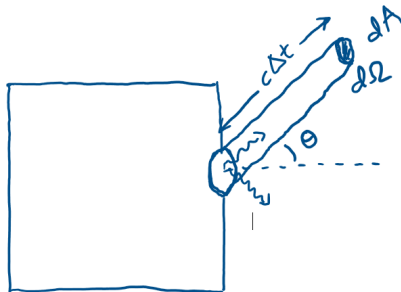
$$\frac{U}{V} = \frac{\pi^2}{15} \frac{(k_B T)^4}{(\hbar c)^3} \quad (34)$$

- The specific heat per unit volume is then

$$c_V = \frac{4\pi^2 k_B^4 T^3}{15(\hbar c)^3} \quad (35)$$

Blackbody radiation

- The radiation spectrum can be measured by making a small hole in the blackbody and measure the radiation that comes out
- The intensity at a particular frequency ω will be the energy coming out per unit area per unit time
- The photons will escape out of an hole of area dA and spread out in a hemisphere



Blackbody radiation

- The energy collected on a hemisphere will be

$$U(\omega) = c\Delta t \int \frac{d\Omega}{4\pi} u(\omega) dA \cos \theta \quad (36)$$

- This gives an intensity of

$$I(\omega) = \frac{c}{4\pi} \int d\Omega u(\omega) \cos \theta = \frac{c}{4} u(\omega) \quad (37)$$

- Then the total energy radiated by the blackbody at a temperature T will be

$$I(T) = \int d\omega I(\omega) = \frac{c}{4} \int d\omega u(\omega) = \sigma T^4 \quad (38)$$

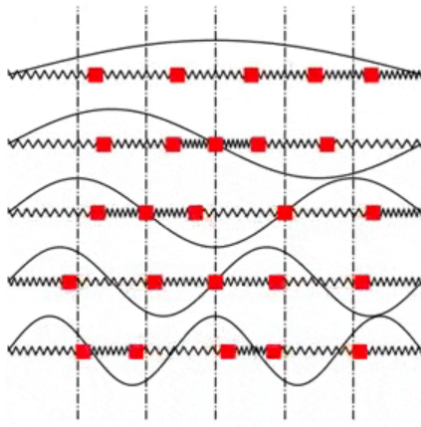
- This is the Stefan's law with the correct σ

$$\sigma = \frac{\pi^2 k_B^4}{60 \hbar^3 c^3} \quad (39)$$

- It is not possible to derive the Stefan's constant classically

Statistics of phonons

- Phonons are vibrations of a lattice



- The frequency of the phonons is

$$\omega = |\mathbf{k}|c_s \quad (40)$$

- where c_s is the speed of sound

Phonon modes

- The waveform of a particular phonon mode is

$$\epsilon e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)} \quad (41)$$

- A phonon has 3 possible polarizations - two transverse and one longitudinal
- There can be any number of phonons in a particular wave mode - so phonons are bosons
- Higher the number of phonons in a particular mode, more is the energy in that mode
- If a solid has N atoms, then there are $3N$ normal modes - assuming each atom independent
- Label the frequencies of these modes

$$\omega_1, \omega_2, \dots, \omega_{3N} \quad (42)$$

- These frequencies will depend on the arrangement of the lattice

Debye model

- In the Debye model, the lowest $3N$ possible frequencies are taken
- As in the case of photons, we will take periodic boundary conditions so the allowed $\mathbf{k}$ values are

$$\mathbf{k} = \frac{2\pi}{L} \mathbf{n}; \quad \mathbf{n} = (n_x, n_y, n_z); \quad n_{x,y,z} = 0, \pm 1, \pm 2, \dots \quad (43)$$

- The number of modes between frequency ω and $\omega + d\omega$ is

$$f(\omega)d\omega = \frac{3V}{2\pi^2 c_s^3} \omega^2 d\omega \quad (44)$$

- The factor of 3 is because of the three polarization modes
- The maximum frequency (ω_m) is obtained by using the Debye model assumption

$$\int_0^{\omega_m} f(\omega) d\omega = 3N \quad (45)$$

Debye model

- This gives

$$\omega_m = c_s \left(\frac{6\pi^2}{v} \right)^{1/3} \quad (46)$$

- where v is defined as the volume per atom

$$v \equiv \left(\frac{V}{N} \right)^{1/3} \quad (47)$$

- The wavelength corresponding this frequency is

$$\lambda_m = \frac{2\pi c_s}{\omega_m} = \left(\frac{4}{3} \pi v \right)^{1/3} \quad (48)$$

- This is approximately equal to twice the inter-particle separation which is correct because wavelengths smaller than interparticle spacing are not physical

Grand partition function

- Each phonon frequency mode is considered as a harmonic oscillator with n_i number of phonons in frequency mode ω_i
- The energy in a particular occupation number configuration $\{n_i\}$ is

$$E\{n_i\} = \sum_{i=1}^{3N} n_i \hbar \omega_i \quad (49)$$

- As for photons, phonons can be spontaneously created by the lattice with very negligible energy- so their chemical potential is negligible
- The grand partition function becomes

$$\mathcal{Z} = \sum_{\{n_i\}} e^{-\beta [\sum_{i=1}^{3N} \hbar \omega_i]} \quad (50)$$

$$= \prod_{i=1}^{3N} \frac{1}{1 - e^{-\beta \hbar \omega_i}} \quad (51)$$

Thermodynamic properties

- Thus we have

$$\ln \mathcal{Z} = - \sum_{i=1}^{3N} \ln(1 - e^{-\beta \hbar \omega_i}) \quad (52)$$

- The average occupation number of the i^{th} mode will be

$$\langle n_i \rangle = - \frac{1}{\beta} \frac{\partial}{\partial (\hbar \omega_i)} \ln \mathcal{Z} = \frac{1}{e^{\beta \hbar \omega_i} - 1} \quad (53)$$

- The internal energy is

$$U = - \frac{\partial}{\partial \beta} \ln \mathcal{Z} = \sum_{i=1}^{3N} \frac{\hbar \omega_i}{e^{\beta \hbar \omega_i} - 1} \quad (54)$$

- This can be converted into the continuum limit

$$U = \int_0^{\omega_m} f(\omega) d\omega \hbar \omega \frac{1}{e^{\beta \hbar \omega} - 1} \quad (55)$$

Thermodynamic properties

- Substituting the expression for density of states $f(\omega)$ we have

$$U = \frac{3V}{2\pi^2 c_s^3} \int_0^{\omega_m} d\omega \omega^2 \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1} \quad (56)$$

- Substitute $t \equiv \beta\hbar\omega$

$$\frac{U}{N} = \frac{9(k_B T)^4}{(\hbar\omega_m)^3} \int_0^{\beta\hbar\omega_m} dt \frac{t^3}{e^t - 1} \quad (57)$$

- Here we define the Debye function $D(x)$ as

$$D(x) \equiv \frac{3}{x^3} \int_0^x dt \frac{t^3}{e^t - 1} \quad (58)$$

- In terms of the Debye function we have

$$\frac{U}{N} = 3k_B T D(\beta\hbar\omega_m) \quad (59)$$

Specific heat

- We define a Debye temperature T_D

$$k_B T_D \equiv \hbar \omega_m = \hbar c \left(\frac{6\pi^2}{v} \right)^{1/3} \quad (60)$$

- And define a $\lambda \equiv T_D/T$ we have

$$\frac{U}{N} = 3k_B T D(\lambda) \quad (61)$$

- One can derive the asymptotic expressions for the Debye function

$$D(x) \approx \begin{cases} 1 - \frac{3}{8}x + \frac{1}{20}x^2 + \dots (x \ll 1) \\ \frac{\pi^4}{5x^3} + \dots (x \gg 1) \end{cases} \quad (62)$$

$$\Rightarrow \frac{U}{N} \approx \begin{cases} 3k_B T \left(1 - \frac{3}{8} \frac{T_D}{T} + \frac{1}{20} \frac{T_D^2}{T^2} + \dots \right); & T \gg T_D \\ 3k_B T \left(\frac{\pi^4}{5} \frac{T^3}{T_D^3} \right) + \dots; & T \ll T_D \end{cases} \quad (63)$$

Specific heat capacity of solids

- The specific heat capacity will becomes

$$\frac{C}{N} = \frac{\partial}{\partial T} \left(\frac{U}{N} \right) \quad (64)$$

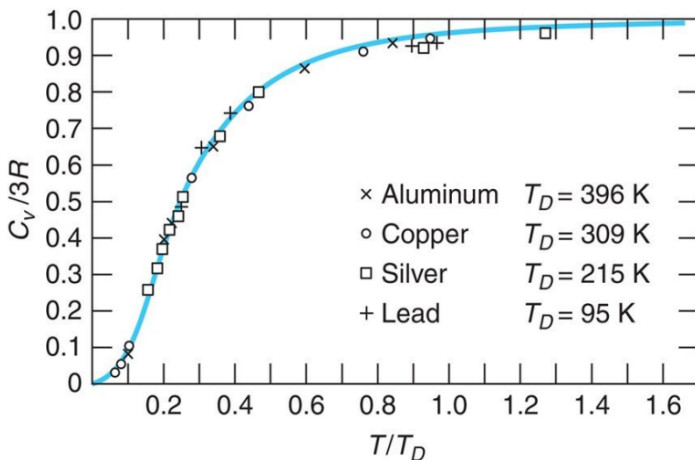
- This gives

$$\frac{C}{Nk} = \begin{cases} 3 \left[1 - \frac{1}{20} \left(\frac{T_D}{T} \right)^2 + \dots \right] & T \gg T_D \\ \frac{12\pi^4}{5} \left(\frac{T}{T_D} \right)^3 + \dots & T \ll T_D \end{cases} \quad (65)$$

- The Debye temperature T_D will depend on different materials
- We already showed that the contribution of electrons (fermions) is negligible to the heat capacity

Experimental results of specific heat

Debye Model of Heat Capacity of Solids



Bose Einstein condensation

- The average number of particles in a grand canonical ensemble is

$$N = - \frac{\partial \ln \mathcal{Z}}{\partial \alpha} \quad (66)$$

- The general form of the grand partition function for a Bose gas is

$$\mathcal{Z} = \prod_k \frac{1}{1 - ze^{-\beta E_k}} \quad (67)$$

$$\Rightarrow \ln \mathcal{Z} = - \sum_k \ln(1 - e^{-\alpha} e^{-\beta E_k}) \quad (68)$$

- From this we get

$$N = \sum_k \frac{ze^{-\beta E_k}}{1 - ze^{-\beta E_k}} \quad (69)$$

- We take the usual case where $\mathbf{k}$ is a wavevector and momentum is $\hbar \mathbf{k}$

Continuum limit

- The summation can be converted into an integral by taking the continuum limit in which the number of states are very large and closely spaced

$$N = \frac{V}{(2\pi)^3} \int_0^\infty dk 4\pi k^2 \frac{ze^{-\beta E_k}}{1 - ze^{-\beta E_k}} \quad (70)$$

$$= V \frac{4\pi}{h^3} \int_0^\infty dp p^2 \frac{ze^{-\beta p^2/2m}}{1 - ze^{-\beta p^2/2m}} \quad (71)$$

- The conversion of summation into integral is valid as long as the different discrete terms in the summation vary smoothly and are finite
- The mode at $\mathbf{k} = 0$ is peculiar because in the conversion from summation to integration, the density of states factor $4\pi p^2 dp$ gives zero weight to the $\mathbf{p} = 0$ mode
- This is fine as long as the contribution of this term is small, which is the case for fermions in which the denominator of this term is $1 + z$

Continuum limit

- However, for bosons, the denominator of $\mathbf{p} = 0$ state becomes $1 - z$ which can become quite small
- For this reason it is better to keep this term separately, while the rest of the terms are still accounted for by the integral

$$N = \frac{4\pi V}{h^3} \int_0^\infty dp p^2 \frac{z e^{-\beta p^2/2m}}{1 - z e^{-\beta p^2/2m}} + \frac{z}{1 - z} \quad (72)$$

- Dividing by V and defining $\nu = V/N$

$$\frac{1}{\nu} = \frac{4\pi}{h^3} \int_0^\infty dp p^2 \frac{1}{z^{-1} e^{\beta p^2/2m} - 1} + \frac{1}{V} \frac{z}{1 - z} \quad (73)$$

- The Bose-Einstein functions $g_\nu(z)$ are defined as

$$g_\nu(z) \equiv \frac{1}{\Gamma(\nu)} \int_0^\infty dx \frac{x^{\nu-1}}{z^{-1} e^x - 1} dx \quad (74)$$

Bose-Einstein functions

- It has been shown that

$$g_\nu(z) = \sum_{l=1}^{\infty} \frac{z^l}{l^\nu} \quad (75)$$

- Making use of this function, we have

$$\frac{1}{\nu} = \frac{1}{\lambda^3} g_{3/2}(z) + \frac{1}{V} \frac{z}{1-z} \quad (76)$$

- Here $\lambda = \sqrt{2\pi\hbar^2/mk_B T}$ is the thermal De Broglie wavelength
- The second term is the contribution to number density from the $\mathbf{p} = 0$ state and the first term is the contribution from the non-zero momentum states
- Represent N_0 as the number of particles in the $\mathbf{p} = 0$ state so we can write

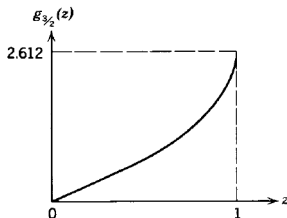
$$\frac{1}{\nu} = \frac{1}{\lambda^3} g_{3/2}(z) + \frac{N_0}{V} \quad (77)$$

Bose-Einstein condensation

- This gives

$$\lambda^3 \frac{N_0}{V} = \frac{\lambda^3}{v} - g_{3/2}(z) \quad (78)$$

- Typically z will be very small and so N_0 will be a negligible fraction of the total number of particles



- The smallest value of N_0 will be given by

$$\frac{\lambda^3}{v} - g_{3/2}(1) \quad (79)$$

Bose Einstein condensation

- When the temperature and the specific volume (v) are such that

$$\frac{\lambda^3}{v} > g_{3/2}(1) \quad (80)$$

- Then a significant fraction of the particles has to be in the $p = 0$ state, which would otherwise be tending to zero
- This phenomenon is called as Bose-Einstein condensation
- It turns out that a system in this state can be treated as having two phases, one having particles with $p > 0$ and another with particles having $p = 0$ which is called the “condensate”
- The phase transition to a condensate occurs at a critical thermal De Broglie wavelength

$$\lambda_c^3 = v g_{3/2}(1) \quad (81)$$

- Qualitatively this means the thermal De Broglie wavelength becomes longer than the average interparticle separation

Bose Einstein condensation

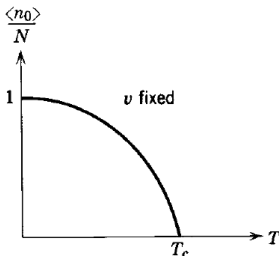
- This gives a critical temperature

$$k_B T_c = \frac{2\pi\hbar^2/m}{[vg_{3/2}(1)]^{2/3}} \quad (82)$$

- Alternatively, for given temperature T we can define a critical specific volume v_c

$$v_c = \frac{\lambda^3}{g_{3/2}(1)} \quad (83)$$

- The equation can be solved numerically to get the correct fraction of particles in the ground state



Experimental observation of BE condensate

- Rubidium atoms were trapped and laser cooled with a critical temperature of 170nK
- The velocity distribution was measured by the spread of the atoms after removal of trap

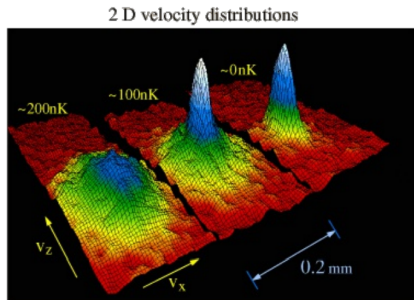


Figure 7. Observation of a Bose-Einstein condensate

The Nobel Prize in Physics 2001



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